



Figure 1: Angstrom login

Here 'sdX' stands for the device file of the SD card. This process will take some time and if it is successful, you will see the cursor jump to the next line and text input will be available. Now remove the SD card.

Booting the Beaglebone

Plug in the SD card into the Beaglebone. To power it up, connect the Beaglebone to your computer via the USB cable provided. Eject the Beaglebone via the Ubuntu disk utility program. I tried ejecting it via the file manager, but that did not work for me. Upon every boot, the Beaglebone is in 'storage mode' by default. Hence, this step is required to switch it to 'network mode'.

The next step is to set up the serial console so that you can interact with the board. There are two possible ways: either a serial console, or the Linux ‘screen’ command. For the first method, install Minicom/Gtkterm (I prefer *minicom*, which you can install with `sudo apt-get install minicom` and run with `sudo minicom -s`). Configure it by selecting the correct USB port, set the baud rate to 115200 and software control, and both the hardware controls to NONE. The other way uses the Linux ‘screen’ command, which is much easier—just run `screen /dev/ttyUSB1 115200` (in most cases the virtual USB serial port is `tyUSB1`; if it does not work, try `tyUSB0`). If all went well, you should be greeted with a familiar Angstrom log-in in the terminal as shown in Figure 1. The username is ‘root’; for the password, just hit ‘Enter’ and you should see the prompt `root@beaglebone:~#` after that.

Toggle the user LED on the Beaglebone

This is the fun part! First, make sure that your kernel supports

GPIO with the following commands in the terminal; the resultant output should have '=v' as shown:

```
$ grep GPIOLIB /boot/config-`uname -r`
CONFIG_ARCH_REQUIRE_GPIOLIB=y
$ grep GPIO_SYSFS /boot/config-`uname -r`
CONFIG_GPIO_SYSFS=y
```

There are four user LEDs on the Beaglebone, accessible from user space on the file system at the location `/sys/class/leds/`. There is one directory per user LED, named as shown below:

```
/sys/class/leds/beaglebone::usr0/  
/sys/class/leds/beaglebone::usr1/  
/sys/class/leds/beaglebone::usr2/  
/sys/class/leds/beaglebone::usr3/
```

In each of these directories, there is a file named *'brightness'*; writing a '1' or a '0' to this file controls the status of that LED, toggling it ON or OFF. Note that since user LED 0 is already used to indicate Ethernet activity, you should use the remaining LEDs for your projects—so I'm going to target LED 3; the following commands will turn it ON and OFF:

```
echo 1 > /sys/class/leds/beaglebone::usr3/brightness
echo 0 > /sys/class/leds/beaglebone::usr3/brightness
```

In the next article in this series, we will do some tinkering and a few mini projects on the Beaglebone. Meanwhile, you can take a look at some amazing hacks/ projects being done by fellow hackers, developers and hobbyists around the world on the Beaglebone, at <http://beagleboard.org/projects/>. **END** 

Resources

While playing around with the Beaglebone, you will have a lot of queries; the following sites would be good places to start looking for answers.

Websites:

- [1] <http://elinux.org/BeagleBone>
- [2] <http://beagleboard.org/bone>
- [3] <http://beagleboardtoys.com/home.html>

Google Group: <https://groups.google.com/forum/?fromgroups#!forum/beagleboard> and on IRC, the #beagle channel.

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